

Governing a Committee of Signals

A pre-registered, 50-year out-of-sample test of SIHRE-style trust and abstention on US industry returns

Paul Russell · independent research · October 2026 · data: Ken French Data Library (49 industry portfolios, 1926–2026)

The question. When you have several forecasting signals that each work some of the time, how should you combine them? The usual answer is to average them equally. SIHRE (Self-Improving Heterogeneous Reasoning Ensemble), the framework I originated, makes two different claims:

- **Trust.** A signal's trust should rise and fall with its recently measured reliability.
- **Abstention.** The system should stay out where its trusted signals disagree.

I tested both claims on five classic equity signals, fixing every choice on 1931–1974 data and locking the plan with a cryptographic hash before looking at 1975–2026.

What I found:

- **Trust works.** Weighting signals by their measured reliability lifted the out-of-sample Sharpe ratio, net of costs, from **−0.02** (equal weight) to **0.30**. The 95% confidence interval for the improvement, **[0.12, 0.58]**, lies above zero. The full SIHRE-lite system also beat equal weight (0.23, CI [0.02, 0.51]), so the primary hypothesis is **supported**.
- **Abstention, as I specified it, does not work.** It roughly doubled turnover and did not reduce drawdowns. That component is **falsified** and is the next thing to redesign.
- **The controls did their job.** A deliberately leaky signal, which peeks at the month it predicts, scored an impossible **Sharpe of 9.7**, and my leakage test flagged it in 25 of 25 months. The five real signals were flagged in 0 of 25.

1. Why this matters, in plain terms

Think of a committee of five advisers. Some are usually right; one, as it turns out, is usually wrong; and who is right changes over the years. Averaging everyone equally gives the bad adviser a full vote. SIHRE instead keeps a running record of how well each adviser's recent calls actually turned out and gives more weight to whoever has been reliable lately, with no one hand-picking the winner in advance. It can also decline to act where trusted advisers contradict each other. This study measures whether those two behaviours are worth anything once trading costs are paid, on data the design never saw.

Real-world setting	The "committee"	What trust and abstention would mean there
Quant equity investing	Alpha signals: momentum, value, quality, revisions, alternative data	Automatically down-weight a signal whose predictive power has decayed, without waiting for a quarterly review; cut exposure where signals conflict. <i>This study.</i>
Retail demand forecasting	Statistical, ML and judgemental forecasts per product and region	Lean on whichever model has recently been accurate for that product; when models disagree sharply, send the forecast to a human planner instead of auto-ordering.
Fraud and credit risk	Rule engines, ML scorers, graph models	Trust each model by its recent precision; route cases where the models disagree to manual review, the costly but correct form of abstaining.
AI and LLM systems	Several models or agents answering the same question	Route to the model that has been reliable for this kind of task; escalate to verification or a person when they disagree.

2. Design

Data and signals

The 49 value-weighted US industry portfolios (monthly, 1926–2026) from the Ken French Data Library. They are public, standard in academic work and free of survivorship bias. There are five "experts", each a classic cross-sectional signal built **only from returns up to the previous month**:

- **MOM:** 12-month momentum, skipping the most recent month.

Ways to combine them

- **EQUAL:** average of the five ranks (the baseline).
- **STATIC:** fixed weights from 1931–1974 reliability.
- **TRUST:** weight each expert by an exponentially weighted average of its monthly information coefficient (IC: the rank correlation between its forecast and what happened), floored at zero.

- **STREV:** 1-month reversal.
- **LTREV:** 3-year reversal.
- **LOWVOL:** low volatility.
- **HIGH52:** nearness to the 12-month high.

Each month every expert ranks the industries; the portfolio is long the top-ranked and short the bottom-ranked (dollar-neutral), net of **10 bps** per unit of turnover.

- **TRUST+ABSTAIN (SIHRE-lite):** TRUST, but take a position in an industry only where trusted experts agree (agreement $\geq \tau$), and go flat when no expert is trusted.
- **EQUAL+ABSTAIN:** an ablation that isolates the abstention rule.

Pre-registration

The data were truncated at 1974 *before* any computation. 24 configurations were compared on 1931–1974 only. The chosen settings (TRUST half-life 60 months; SIHRE-lite half-life 12, $\tau = 0.75$), four hypotheses and the test rules were written to a file and SHA-256-hashed (`fb49fac5...`). The holdout script refuses to run if that file changes. It ran once.

3. Results on the 1975–2026 holdout (620 months)

Strategy (net of 10 bps costs)	Sharpe	Return / yr	Volatility	Max drawdown	Turnover / mo	Sharpe at 0 / 25 bps
EQUAL (baseline)	−0.02	−0.2%	11.0%	−59.7%	1.26	0.12 / −0.22
STATIC (calibration weights)	0.31	3.9%	12.6%	−41.7%	0.78	0.38 / 0.19
TRUST	0.30	3.8%	12.8%	−43.8%	0.82	0.38 / 0.18
TRUST+ABSTAIN (SIHRE-lite)	0.23	3.3%	14.4%	−47.8%	1.43	0.35 / 0.05
EQUAL+ABSTAIN (ablation)	−0.11	−1.8%	17.0%	−87.2%	2.78	0.09 / −0.40
Momentum alone (best single expert, also best in calibration)	0.38	4.6%	12.2%	−43.6%	0.73	0.45 / 0.27
1-month reversal alone (the "bad adviser")	−0.73	−7.3%	10.0%	−98.5%	2.56	−0.42 / −1.19
Control: placebo (random scores)	−0.58	−3.3%	5.7%	−84.1%	2.66	−0.02 / −1.42
Control: leaky (peeks at the predicted month)	9.66	110%	11.4%	0.0%	2.56	9.94 / 9.25

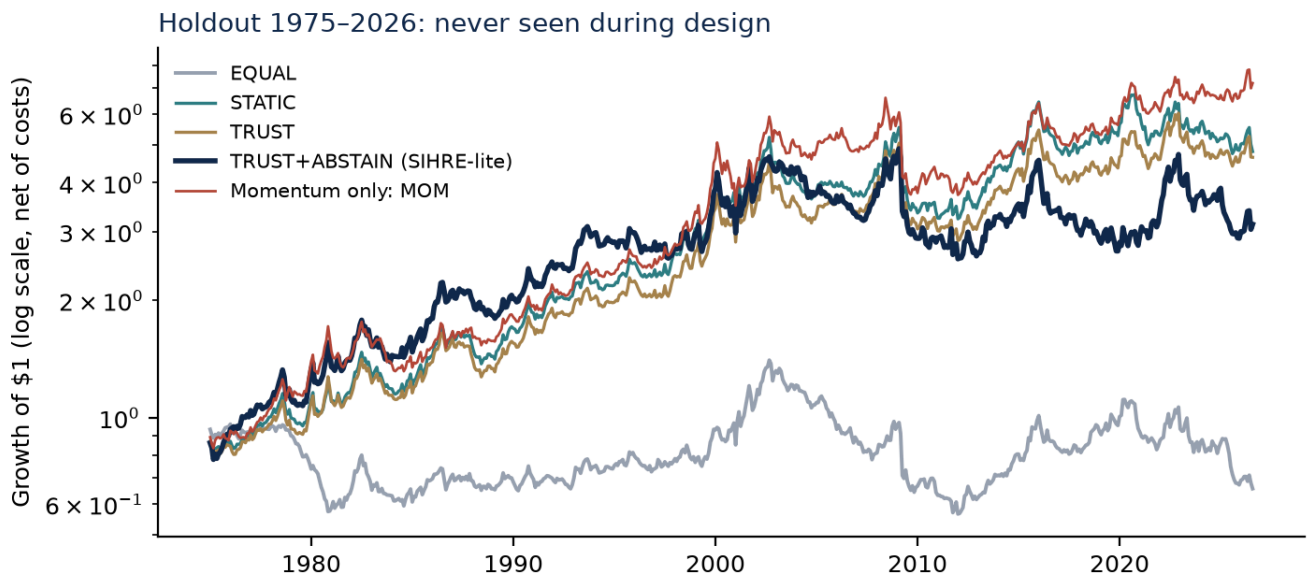


Figure 1. Growth of \$1 on the holdout, net of costs, log scale. Equal weighting loses money; any form of reliability weighting does not.

Pre-registered hypothesis (holdout)	Sharpe difference	95% CI*	Verdict
H1 (primary): SIHRE-lite beats EQUAL	+0.25	[+0.02, +0.51]	Supported
H2: TRUST beats EQUAL	+0.32	[+0.12, +0.58]	Supported
H3: abstention lowers max drawdown (SIHRE-lite vs TRUST)	−47.8% vs −43.8%	—	Not supported
H4: placebo's Sharpe CI includes 0	−0.58	[−0.91, −0.29]	Failed as written†
Ablation: SIHRE-lite vs TRUST (does abstention add value?)	−0.07	[−0.23, +0.06]	No evidence it helps
Ablation: SIHRE-lite vs STATIC	−0.08	[−0.26, +0.07]	No significant difference

* Moving-block bootstrap, 12-month blocks, 5,000 resamples, paired months. † The placebo has no skill (gross Sharpe −0.02) but trades a lot, so a *net-of-cost* test was bound to reject it. My pre-registered test should have used gross returns. I report the registered result unchanged and flag the design error rather than re-running.

4. What the results mean

- **Most of the value comes from not trusting a bad signal.** One-month reversal, a strong effect in individual stocks, *loses* money between industries: industries show short-term momentum, not reversal, a known result (Moskowitz & Grinblatt, 1999). Equal weighting gives it a full 20% vote. Reliability weighting learned on its own to cut it to about 9% on average (Figure 2). Net Sharpe moved from −0.02 to 0.30.
- **Adaptive trust matched a well-chosen fixed weighting; it did not beat it.** Over this 50-year holdout, weights fixed from 1931–1974 did as well as weights that adapted (0.31 vs 0.30). Adaptive trust earns its place when there is no long history to calibrate on, or when a signal's power decays. Figure 2 shows trust leaving LOWVOL in the 2000s and moving towards LTREV in the 2020s.
- **Abstention, as specified, failed, and the evidence shows why.** Dropping industries where experts disagreed concentrated the book and churned it: turnover rose from 0.82 to 1.43 a month, volatility rose, and costs ate the benefit. At zero cost SIHRE-lite's Sharpe is 0.35; at 25 bps it is 0.05. The idea that disagreement should lower conviction is untouched; this way of acting on it is rejected.
- **No combination protected against the 2009 momentum crash.** In March–May 2009 every combination lost 28–34% (momentum alone lost 32%; only the two reversal signals held up). A trust estimate with a 12- to 60-month memory is too slow for a three-month regime break.
- **The best single signal, momentum, won outright** (0.38), and it was also the strongest signal in the 1931–1974 calibration (0.42; next best 0.14). A simple rule of picking the best calibration signal would therefore have beaten every combination here. The combination's case rests on settings this test does not cover: many signals, short histories, or a leading signal that decays.

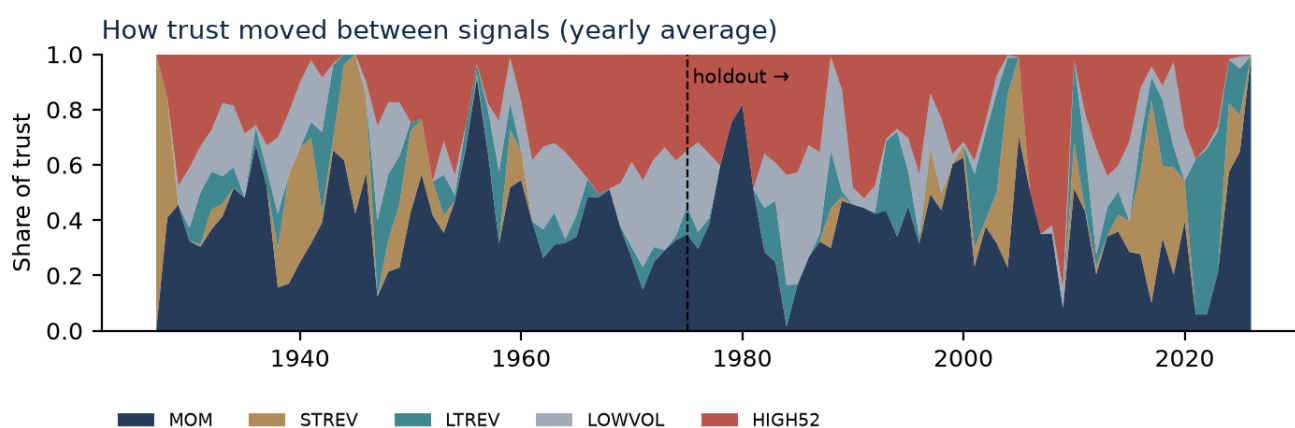


Figure 2. How SIHRE-lite's trust moved between the five signals (yearly average). STREV, the signal that loses money between industries, averages about 9% of trust, against 20% under equal weighting.

5. What the controls show

- **Leakage test.** I perturbed each test month's returns, and every later month's, and checked whether the signal for that month changed. Clean signals: changed in 0 of 25 months. The deliberately leaky signal: 25 of 25. The instrument detects what it is meant to detect, and does not raise false alarms.
- **Why it matters.** The leaky signal's Sharpe of 9.7 is what one look-ahead bug looks like. An earlier ML backtest of mine (2024) had four such flaws: a shuffled train/test split, same-day features, scaling fitted before the split, and in-sample predictions. This study is built so that class of error is tested for, not assumed away.
- **Multiple testing.** 24 configurations were tried, all before the holdout; the holdout tested two primary hypotheses.

7. Reproducibility

- **Data hashes:** industries zip 16b3f20b..., factors zip 593f4fbe....
- **Pre-registration:** results/preregistration.json, SHA-256 fb49fac5....
- **To reproduce:** `python src/run_holdout.py` verifies the stored hash, then reruns the holdout. `python src/run_calibration.py` re-derives the same chosen parameters; its timestamp, and so its hash, will differ, so run it in a copy of the folder.
- **Outputs:** all figures above come from results/holdout_results.json.

6. Limits and next steps

- **Industries, not stocks.** 49 portfolios is a small cross-section. Next: individual stocks with point-in-time constituents.
- **Simple costs.** A flat 10 bps per unit of turnover, with no market impact or capacity.
- **Abstention v2.** Scale *total exposure* by agreement rather than dropping positions, with a turnover penalty, pre-registered before the next holdout.
- **Faster trust under stress.** Add a short-memory trust signal or a regime detector for crash months like 2009.
- **Correct the placebo test** to use gross returns in the next pre-registration.